SEIZELUT: A LUT-NATIVE MULTIMODAL SEIZURE DETECTOR FOR WEARABLE DEVICES ON FPGA

Phase I: Dataset Study, Baselines and LUT Network Design

Phase I Project Progress Report submitted to the
National Institute of Technology Warangal
in partial fulfilment of the requirements for the award of the degree

of

Bachelor of Technology
in Electronics and Communication Engineering

by

Sai Chaithr (Roll No. 23ECB0A20)

Under the guidance of

[Guide's name with title]
[Designation], Department of Electronics and Communication Engineering



DEPARTMENT OF ELECTRONICS AND COMMUNICATION ENGINEERING
NATIONAL INSTITUTE OF TECHNOLOGY WARANGAL
November 2026

APPROVAL OF THE PROJECT EVALUATION BOARD

DD/MM/YYYY

Certified that the report entitled **SeizeLUT: A LUT-Native Multimodal Seizure Detector for Wearable Devices on FPGA** submitted by Sai Chaithr (Roll No. 23ECB0A20) to the National Institute of Technology Warangal, as the Phase I progress report of the Bachelor of Technology project in Electronics and Communication Engineering, has been examined and accepted by the evaluation board.

(Member of the Board) (Member of the Board) (Member of the Board)

(Supervisor) (Co-Supervisor)

(Head of the Department) (Chairperson, Evaluation Board)

(The certificate is to be printed on the Departmental Letter-Head)

CERTIFICATE

This is to certify that the report entitled **SeizeLUT: A LUT-Native Multimodal Seizure Detector for Wearable Devices on FPGA**, submitted by **Sai Chaithr (Roll No. 23ECB0A20)** to the National Institute of Technology Warangal, is a record of bona fide project work carried out by the student under my (our) supervision during Phase I of the Bachelor of Technology project in Electronics and Communication Engineering.

_____

Supervisor
[Guide's name with title]

Date:

DECLARATION

I certify that

a. The work contained in the report is original and has been done by me under the general supervision of my supervisor(s).

b. The work has not been submitted to any other Institute for any degree or diploma.

c. I have followed the guidelines provided by the Institute in writing the report.

d. Whenever I have used materials (data, theoretical analysis, and text) from other sources, I have given due credit to them by citing them in the text of the report and giving their details in the references.

e. Whenever I have quoted written materials from other sources, I have put them under quotation marks and given due credit to the sources by citing them and giving required details in the references.

f. An AI assistant was used to help write analysis code, check results and draft parts of the text. All numbers in this report come from my saved experiment records and scripts, and I take responsibility for the content.

Sai Chaithr

(23ECB0A20)

Signature of the Student

Acknowledgements

I thank my project guide, [Guide's name with title], [Designation], Department of Electronics and Communication Engineering, for the guidance and encouragement given throughout Phase I of this project.

I thank the Head of the Department and the faculty of the Department of Electronics and Communication Engineering, National Institute of Technology Warangal, for the facilities and support provided. I also thank the SeizeIT2 consortium for releasing the dataset used in this work publicly, and the authors of the open-source tools on which my software builds.

Finally, I thank my family and friends for their support.

Sai Chaithr

Abstract

A wearable seizure detector has to run for days on a small battery. We ask whether a *LUT-native* neural network, whose neurons are the six-input lookup tables (LUTs) of an FPGA, can detect seizures from behind-the-ear EEG, ECG, EMG and motion in unseen patients, with no multipliers and a few thousand LUTs.

On SeizeIT2 (125 patients, 11,626 hours, 883 seizures; official split of 96 training and 29 test patients) we audited the dataset, reproduced the published ChronoNet baseline (and found that its code tunes on the test patients), built a bit-exact integer pipeline of 117 features with no accuracy loss, and compared nine model families with patient-wise cross-validation and SzCORE event metrics. Interpretability of the network, LightGBM-guided feature selection and distillation from LightGBM gave the final design.

Scored once on the 29 held-out test patients (5 seeds), the final network reaches an event-F1 of 0.248 with 0.8 false alarms per day at about 900 LUTs and no multipliers: significantly better than a clean ChronoNet (0.027; $p = 0.027$), above TreeLUT (0.233) and an INT8 network (0.155), level with random forests (0.246) and below LightGBM (0.308). At the base paper's own metric it detects 43.2 % of seizures at 1.24 false alarms per hour, against 34.8 % for ChronoNet at the same rate; with a pre-registered post-ictal verification stage (a second LUT network) it detects 73.3 % at 10.9 false alarms per hour, above the published SVM (71.1 % at 11.0). Its Verilog matches the software on all 431,960 windows of 10 test recordings. Phase II will implement and measure it on an FPGA.

Index Terms: epilepsy, seizure detection, wearable EEG, behind-the-ear EEG, multimodal fusion, weightless neural networks, lookup tables, FPGA, SeizeIT2.

Contents

Certificate of Approval ii

Certificate iii

Declaration iv

Acknowledgements v

Abstract vi

Contents vii

List of Figures ix

List of Tables xi

List of Abbreviations xiii

1 Introduction 1

1.1 Motivation 1

1.2 Problem Statement and Objectives 1

1.3 Contributions 2

1.4 Organisation of the Report 2

2 Literature Review 3

2.1 Epilepsy and Wearable Seizure Detection 3

2.2 Detection Algorithms and Evaluation 3

2.3 LUT-Native, Weightless and Logic Networks 5

2.4 Hardware Seizure Detectors 6

2.5 Research Gaps 8

3 Dataset, Methodology and Experimental Setup 9

3.1 SeizeIT2 Dataset 9

3.2 Audit Findings 11

3.3 System Architecture 11

3.4 The LUT-Native Classifier (DWN) 12

3.5 Models and Extensions 13

3.6 Evaluation Protocol 13

4 Experimental Results and Comparisons 15

4.1 Reproduction of the Published Baseline 15

4.2 Integer Versus Floating-Point Features 16

4.3 Comparison of Model Families 16

4.4 What Improved the Network 18

4.5 Missing Sensors 19

4.6 Seizure Types 20

4.7 Final Evaluation on the Held-Out Test Patients 20

4.8 Comparison with the Base Paper 22

4.9 Post-Ictal Verification (Secondary Analysis) 23

4.10 Hardware Handoff Check 24

5 Challenges Faced and How They Were Addressed 25

6 Discussion, Conclusions and Future Scope 27

6.1 Key Findings 27

6.2 Status at the End of Phase I 28

6.3 Plan for Phase II 29

6.4 Conclusions 29

References 30

A Full Results and Sources 36

A.1 All Models on the 29 Test Patients 36

A.2 Sources of the Reported Numbers 36

List of Figures

3.1 Seizure classes in the training and test patients: 822 of 883 seizures are focal seizures that do not become tonic-clonic. 10

3.2 Hours recorded (bars; blue: training, orange: test) and seizures per patient (dots). 10

3.3 The proposed detector. Blue: bit-exact integer front end; orange: LUT-native classifier. 11

3.4 Estimated hardware cost. Left: the final DWN by block (the encoder dominates). Right: total LUTs of each model, log scale. 12

4.1 ChronoNet on the 29 test patients: published and reproduced (the paper's metrics). 15

4.2 Event-F1 against estimated LUTs for every LUT-native run (training folds), with the Pareto front. 17

4.3 Every extension against the base DWN: mean change in per-patient event-F1 with 95 % interval. *Modality dropout was adopted for its robustness to missing sensors (Section 4.5). 18

4.4 N7: inputs read by the firing seizure LUTs during true and false alarms (first, mutual-information network): false alarms look like seizures to it. 19

4.5 Event-F1 with sensors removed at test time (training folds). Distillation is the most accurate with all sensors; modality dropout keeps working without ECG or motion. 19

4.6 Seizures detected by type (base DWN, training folds): tonic-clonic seizures are caught, focal seizures rarely. 20

4.7 Event-F1 on the 29 test patients (mean $\pm$ std over 5 seeds) and false alarms per day. 20

4.8 Accuracy against hardware cost on the test patients: the LUT networks reach tree-ensemble accuracy at a fraction of the LUTs. 21

4.9 Seizures detected against false alarms per hour on the 29 test patients (the base paper's metrics). Curves: threshold sweeps of the frozen models (mean of 5 seeds; band: std of our network), drawn as far as every seed's sweep reaches; dots: our two operating points, chosen on the training folds; stars: the two-stage detector (Section 4.9); open markers: published values. 22

List of Tables

1.1 Objectives and their status at the end of Phase I. 2

2.1 Wearable and multimodal seizure detection studies (results as reported). 4
2.2 Seizure detection algorithms and evaluation frameworks (results as reported). . 5
2.3 LUT-native, weightless and logic networks (results as reported). 6
2.4 Hardware seizure detectors (results as reported; units differ between papers). . . 7
2.5 Research gaps and how this project addresses them. 8

3.1 Signals recorded by the wearable (rates measured from the files). 9
3.2 Inventory of SeizeIT2 measured by our audit. 10
3.3 Where the released files differ from the documentation, and what we did. . . . 11
3.4 The 117 integer features (per 2-s window; many also as change from a 64-s moving average). 12
3.5 Models compared on the same features, splits and metrics. 13
3.6 Extensions of the DWN tested in this project (adopted only if significantly better). 13

4.1 The integer front end against its floating-point twin (10 training patients). . . . 16
4.2 Five-fold cross-validation on the 96 training patients (best-F1 operating point). 16
4.3 The changes that passed the significance test. 18
4.4 Final DWN against the main references on the 29 test patients. 21
4.5 The base paper's baselines and our network on the same 29 test patients (the paper's metrics). 23
4.6 Two-stage detector with post-ictal verification. 24
4.7 Verilog of the final DWN against the software on 10 test recordings. 24

5.1 The main problems of Phase I and their solutions. 25

6.1 Status of the project activities. 28

6.2 Work carried out during Phase I. 28
6.3 Month-wise plan for Phase II. 29

A.1 Final evaluation of every model on the 29 test patients. 36
A.2 Sources of the reported numbers. 37

List of Abbreviations

ACC	Accelerometer
ASIC	Application-specific integrated circuit
AUPR	Area under the precision–recall curve
AUROC	Area under the receiver operating characteristic curve
bte-EEG	Behind-the-ear electroencephalography
CNN	Convolutional neural network
CV	Cross-validation
DSP	Digital signal processing block (FPGA multiplier)
DWN	Differentiable Weightless Neural Network
ECG	Electrocardiography
EDF	European Data Format
EFD	Extended Finite Difference
EMG	Electromyography
FA	Focal aware (seizure); false alarm
FBTC	Focal to bilateral tonic-clonic (seizure)
FIA	Focal impaired awareness (seizure)
FP/24 h	False positives (false alarms) per 24 hours
FPGA	Field-programmable gate array
GBDT	Gradient-boosted decision trees
GRU	Gated recurrent unit
GYR	Gyroscope
iEEG	Intracranial EEG
ILAE	International League Against Epilepsy
INT8	8-bit integer format
LGN	Logic-gate network
LSB	Least significant bit

LUT	Lookup table (LUT6: six-input lookup table)
MCU	Microcontroller
MI	Mutual information
MIL	Multiple-instance learning
MLP	Multilayer perceptron
OVLP	Any-overlap event scoring
SoC	System on chip
SUDEP	Sudden unexpected death in epilepsy
SVM	Support vector machine
SzCORE	Seizure Community Open-Source Research Evaluation framework
TCN	Temporal convolutional network
TUSZ	Temple University Hospital Seizure Corpus
WNN	Weightless neural network

Chapter 1

Introduction

1.1 Motivation

- Epilepsy affected an estimated 45.9 million people in 2016 [1]. Patients often miss their own seizures, so diaries undercount them [2], and tonic-clonic seizures carry a risk of sudden death [3].
- Wearables are recommended today only for tonic-clonic seizures [4]; for focal seizures (822 of the 883 in SeizeIT2) false alarms are the main obstacle [5].
- Behind-the-ear EEG with ECG, EMG and motion is a discreet alternative [6,7]; SeizeIT2 [8] provides 125 patients recorded with such a wearable.
- A wearable detector must run for days on a small battery. Multiplications dominate the cost of neural networks in hardware, while an FPGA's basic element is a 6-input lookup table (LUT).
- *LUT-native* networks [9–11] are built from exactly this element and need no multipliers; they have never been used for seizure detection.

1.2 Problem Statement and Objectives

Can a LUT-native neural network, fed by integer features from behind-the-ear EEG, ECG, EMG and motion, detect seizures in unseen patients as well as the published SeizeIT2 baselines, with no multipliers and a few thousand LUTs?

Table 1.1 Objectives and their status at the end of Phase I.

#	Objective	Status
1	Audit the SeizeIT2 dataset and reproduce its published baselines	Done
2	Integer-only, bit-exact feature pipeline with no accuracy loss	Done
3	Compare LUT-native, tree and multiplier-based models on accuracy and hardware cost	Done
4	Extend the LUT network for wearables (memory, encoders, distillation, missing sensors)	Done
5	Evaluate the frozen design once on the 29 held-out test patients	Done
6	Verified Verilog handoff; FPGA implementation	Handoff done; FPGA in Phase II

1.3 Contributions

- The first differentiable LUT-native seizure detector, and the first benchmark of LUT-family models on SeizeIT2.
- A bit-exact integer multimodal front end with a fixed-point specification.
- Distillation from a tree ensemble into a LUT network (fewer false alarms) and modality dropout (robustness to a missing ECG).
- An independent reproduction of the SeizeIT2 ChronoNet baseline, including a protocol with no test information.
- A hardware handoff whose Verilog matches the software on 431,960 test windows.

1.4 Organisation of the Report

Chapter 2 reviews the literature, Chapter 3 the dataset and method, Chapter 4 the results, Chapter 5 the challenges, and Chapter 6 the conclusions and the plan for Phase II.

Chapter 2

Literature Review

The survey covers 79 references in four areas. Each was checked against its published record; numbers are as reported in the cited abstracts.

2.1 Epilepsy and Wearable Seizure Detection

- **Clinical need.** Seizures are classified by onset and awareness [12]; unaware patients under-report them [2]; the ILAE–IFCN guideline recommends wearables only for tonic-clonic seizures [4].
- **Devices.** EMG, wristband and multimodal devices detect convulsive seizures well [13–17]; focal seizures remain open.
- **Behind-the-ear EEG (SeizeIT).** Two behind-the-ear channels detect seizures [6]; ECG adds 8–11 % sensitivity [7, 18]; tree ensembles won the 2023 challenge [19]; SeizeIT2 released 125 patients [8]; a 192-adult validation reached precision 0.004 before review [5]; ECG-only [20] and complex-valued multimodal [21] detectors also use SeizeIT2.
- **Fusion.** Adding EMG and motion mainly removes false alarms [22]; only one work handles a missing modality [23].

2.2 Detection Algorithms and Evaluation

- **Feature-based models** work well but are often patient-specific [25]; tree ensembles are strongest on SeizeIT data [26, 27].
- **Deep models** (ChronoNet [28], EEGNet [29], EEGWaveNet [30], graph and transformer models [31–33]) are large and multiplier-based; ChronoNet trained on SeizeIT1 reached

Table 2.1 Wearable and multimodal seizure detection studies (results as reported).

Work	Year	Signals	Seizures	Result	Gap for our problem
Beniczky et al. [14]	2018	EMG	tonic-clonic	>90 % sensitivity	convulsive only
Onorati et al. [15]	2017	EDA + ACC	convulsive	94.55 %, 0.2 FA/day	convulsive only
Naganur et al. [17]	2022	meta-analysis	tonic-clonic	0.91, 2.1 FA/24 h	focal not solved
Vandecasteele et al. [6]	2020	bte-EEG	focal	64.1 %, 2.8 FP/24 h (patient-independent)	EEG only
Vandecasteele et al. [7]	2021	bte-EEG + ECG	focal	+8–11 % sensitivity with ECG	software only
Bhagubai et al. [19]	2024	bte-EEG	challenge	tree ensemble won	no hardware budget
Zhang et al. [22]	2024	EEG + EMG + ACC	tonic-clonic	90.9 %, 0.1 FP/24 h	tonic-clonic only
Swinnen et al. [5]	2026	SeizeIT2 wearable	focal, 192 adults	0.73 at precision 0.004	false alarms
BrainFuseNet [24]	2024	EEG + PPG + ACC	–	64.28 %, 0.21 FP/h, MCU	multipliers, no ECG/EMG
Zhang et al. [23]	2024	EEG + EMG	–	–	large transformer

15.2 % sensitivity on SeizeIT2 [27].

- **Imperfect labels**: weak supervision, MIL, noisy-label training and distillation [34–37] motivate our extensions N4 and N5.
- **Evaluation** varies between papers and corpora [38, 39]; SzCORE standardises event metrics [40]; in a blind benchmark the best event-F1 was 43 % [41]; subject leakage inflates results [42].

Table 2.2 Seizure detection algorithms and evaluation frameworks (results as reported).

Work	Year	Method	Data	Result	Gap for our problem
Shoeb & Guttag [25]	2010	patient-specific SVM	CHB-MIT	96 % of seizures, 2 FD/24 h	patient-specific, scalp
SeizFt [26]	2023	features + CatBoost	SeizeIT1/2	ICASSP 2023 winner	EEG only, no hardware cost
Huang et al. [27]	2023	random forest; ChronoNet	SeizeIT1/2	RF 73.6 %; ChronoNet 15.2 %	EEG only
ChronoNet [28]	2019	CNN + GRU	TUH abnormal	benchmark at the time	large, multipliers
EEGWaveNet [30]	2022	multiscale CNN	CHB-MIT, TUSZ	better subject-independent	scalp, float CNN
Tang et al. [31]	2022	self-supervised GNN	TUSZ	AUROC 0.875	needs full montage
Shama & Venkataraman [36]	2026	noisy-label training	TUH, CHB-MIT	robust to label noise	scalp EEG
Zheng et al. [37]	2023	distilled GNN	TUSZ	competitive with 2 channels	float student
SzCORE [40]	2024	evaluation framework	scalp EEG	standard event metrics	not for wearables
SzCORE challenge [41]	2025	blind benchmark	65 subjects	best event-F1 43 %	scalp EEG

2.3 LUT-Native, Weightless and Logic Networks

- **Weightless networks** replace weighted sums with memory lookups [43–45]; the **DWN** made LUT contents and input mapping trainable by gradient descent [9], later applied to activity recognition and control [46, 47].
- **Logic-gate networks** learn 2-input gates [10, 48, 49]; a recurrent version exists for language [50].

- **LUT networks for FPGAs**: LUTNet, LogicNets, PolyLUT, NeuraLUT, AmigoLUT, ReducedLUT and TreeLUT [11, 51–56].
- **The encoder** can dominate the cost of small DWNs [57]; in our final network it is 776 of 895 LUTs.
- **Biosignals**: ECG [58] and EEG classification [59]; the only weightless seizure work is a 4-patient study [60].

Table 2.3 LUT-native, weightless and logic networks (results as reported).

Work	Year	Model	Domain	Hardware / result	Gap
WiSARD [43]	1984	RAM-node WNN	images	–	no gradient training
ULEEN [45]	2023	weightless	MLPerf Tiny	7.1× energy eff. vs FINN	no biosignals
DWN [9]	2024	differentiable LUT6 net	tabular	FPGA, MCU	stateless, no events
nanoML HAR [46]	2025	DWN	inertial HAR	56–104 nJ / sample	single modality
LGN [10]	2022	2-input logic gates	images	>1 M images/s (CPU)	no biosignals
Recurrent LGN [50]	2025	stateful logic gates	translation	–	no hardware
LogicNets [11]	2020	sparse quantised LUT net	jet tagging	<15 ns latency	tiny fan-in
NeuraLUT [53]	2024	dense sub-nets in LUTs	jet tagging	4.3× lower latency	static
TreeLUT [56]	2025	quantised GBDT	tabular	4–101× lower area-delay	per window
Mecik & Kumm [57]	2025	DWN generator	jet tagging	encoder up to 3.2× LUTs	–
Mommen et al. [58]	2026	LGN, LUT network	ECG arrhythmia	2.0–3.0k LUTs, 5–7 mW	single modality
Dharia et al. [59]	2026	LGN	EEG	embedded CPU	no FPGA, no seizures
de Aguiar et al. [60]	2015	WiSARD	seizure anticipation	–	4 patients, accuracy only

2.4 Hardware Seizure Detectors

- **ASICs** reach nJ–µJ per decision with SVMs and trees [61–66].

- **FPGAs**: SVM, line-length and spiking-network detectors [67–71]; the closest to ours is an SNN on a small iCE40 [71].
- **Microcontrollers**: e-Glass, EEGformer, BrainFuseNet [24, 72, 73] and low-power EEG platforms [74, 75]; efficient-inference tools: BNNs, FINN, INT8 quantisation, hls4ml [76–79].
- Nearly all use EEG only, per-patient training and epoch-level metrics; none maps the classifier into LUTs.

Table 2.4 Hardware seizure detectors (results as reported; units differ between papers).

Work	Year	Platform	Signals	Hardware	Detection	Gap
Verma et al. [61]	2010	ASIC	scalp EEG	9 µJ / feature vector	classifier off-chip	EEG only
Bin Altaf et al. [62]	2015	ASIC	scalp EEG	2.73 µJ / class.	95.7 % (epoch)	per patient
Shoaran et al. [65]	2018	ASIC	iEEG	41.2 nJ / class.	F1 87.9 %	implant
NeuralTree [66]	2022	ASIC	EEG, iEEG	0.227 µJ / class.	95.6 % (EEG)	neural only
Wang et al. [67]	2018	FPGA (Zynq)	scalp EEG	380 mW	98.4 %, 0.356 FD/h	per patient
Farhang Razi & Schmid [69]	2022	FPGA (Cyclone V)	iEEG	0.49 mW dynamic	100 % sens.	implant
Busia et al. [71]	2025	FPGA (iCE40)	scalp EEG	4.55 nJ / inference	0.3 FP/h	EEG only
e-Glass [72]	2018	MCU	scalp EEG	2.71 days / charge	93.8 % (epoch)	EEG only
EEGformer [73]	2024	MCU	scalp EEG	0.31 mJ / inference	73 %, 0.15 FP/h	multipliers
BrainFuseNet [24]	2024	MCU	EEG, PPG, ACC	0.11 mJ / inference	0.21 FP/h	no ECG/EMG

2.5 Research Gaps

Table 2.5 Research gaps and how this project addresses them.

#	Gap	Evidence	How this project addresses it
1	No LUT-native (weightless, logic-gate or LUT) seizure detector exists	LUT networks are benchmarked on images, physics and tabular data; the only weightless seizure work is a 4-patient study [60]	First benchmark of DWN, LGN, TreeLUT and LogicNets-style networks for seizure detection, on SeizeIT2
2	Hardware detectors use EEG alone, although fusion is the most consistent way to cut false alarms	[7, 22]; Section 2.4	Integer features from behind-the-ear EEG, ECG, EMG and motion in one bit-exact front end
3	Missing sensors are rarely handled; LUT networks never	[23]; Section 2.3	Presence bits, modality dropout, gated and late fusion, missing-sensor tests (N6)
4	The encoder's hardware cost is ignored, although it dominates small LUT networks	[57]; our estimate: about 850 of 960 LUTs	Cost model that counts every comparator; comparator-budgeted and learned encoders (N2)
5	LUT networks are stateless; seizures unfold over tens of seconds	[50]	A stateful DWN with state bits fed back every second (N1)
6	Results are patient-specific, leak between training and test, or use incomparable metrics	[38, 41, 42]	Patient-wise cross-validation, a locked test set used once, SzCORE metrics with the paper's metrics alongside
7	Small models learn from noisy labels and from larger models only in floating point	[36, 37]	Noise-robust and multiple-instance losses (N5) and distillation from LightGBM (N4) into a LUT network
8	Detectors are black boxes	Section 2.3	Tracing alarms to the inputs read by each LUT (N7)
9	Published wearable baselines are not independently reproduced	Section 2.1	Independent reproduction of the SeizeIT2 ChronoNet baseline, with and without test-patient information

Chapter 3

Dataset, Methodology and Experimental Setup

3.1 SeizeIT2 Dataset

SeizeIT2 [8] (OpenNeuro ds005873 v1.1.0, CC0) holds wearable recordings of 125 patients with focal epilepsy from five European epilepsy units, with seizures annotated on hospital video-EEG. We use the official split: **96 training patients** for all development and **29 test patients** scored once, after the design was frozen. Tables 3.1 and 3.2 and Figures 3.1 and 3.2 summarise it (all values measured by our audit).

Table 3.1 Signals recorded by the wearable (rates measured from the files).

Signal	Channels	Rate
Behind-the-ear EEG	2 (left, right or cross-head)	256 Hz
ECG	1	256 Hz
EMG	1	256 Hz
Motion	3-axis accelerometer + 3-axis gyroscope	25 Hz

Table 3.2 Inventory of SeizeIT2 measured by our audit.

Quantity	**Value**
Patients / recordings / EDF files	125 / 2,850 / 11,009 (125.9 GB)
Hours of EEG	11,626 h (median 95 h per patient)
Seizures	883 (train 702, test 181)
Seizure time	14.3 h = **0.123 %** of the recording time
Seizure duration	2–962 s (median 33 s)
Missing ECG/EMG; missing motion	3.1 %; 3.9 % of the time

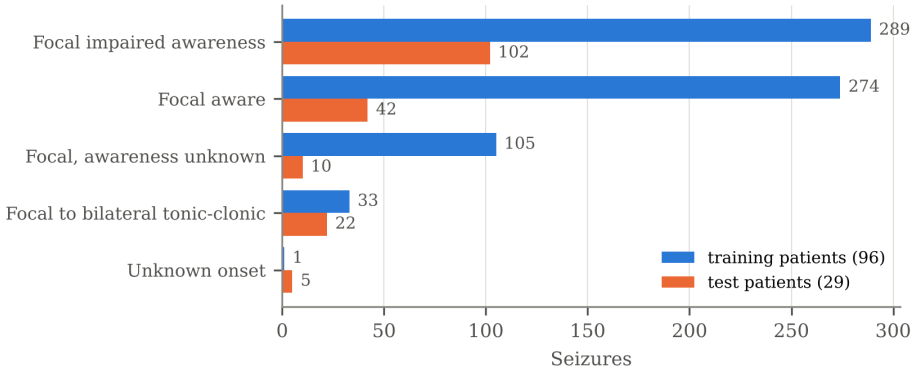


Figure 3.1 Seizure classes in the training and test patients: 822 of 883 seizures are focal seizures that do not become tonic-clonic.

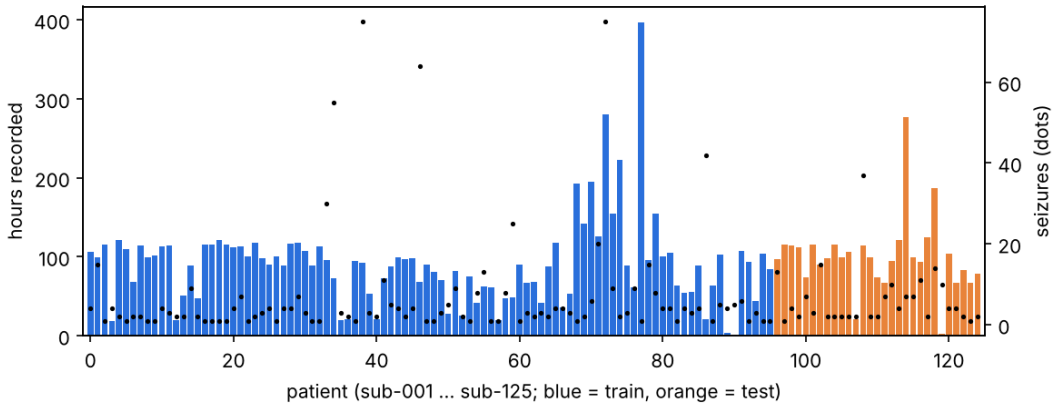


Figure 3.2 Hours recorded (bars; blue: training, orange: test) and seizures per patient (dots).

3.2 Audit Findings

Table 3.3 Where the released files differ from the documentation, and what we did.

Item	Documentation	Files	Action
Sampling rate	250 Hz	256 Hz	Filters designed for 256 Hz
Seizures	886	883	Report our count
Centre and age labels	used for splits	not released	Folds by seizure count
Official artefact rule	13–150 µV	flags 69 % of normal windows	Flags used as input bits
Downloads	–	9 files cut short	Size checks; re-extracted
Evaluation code	–	two defects	Corrected values reported alongside

3.3 System Architecture

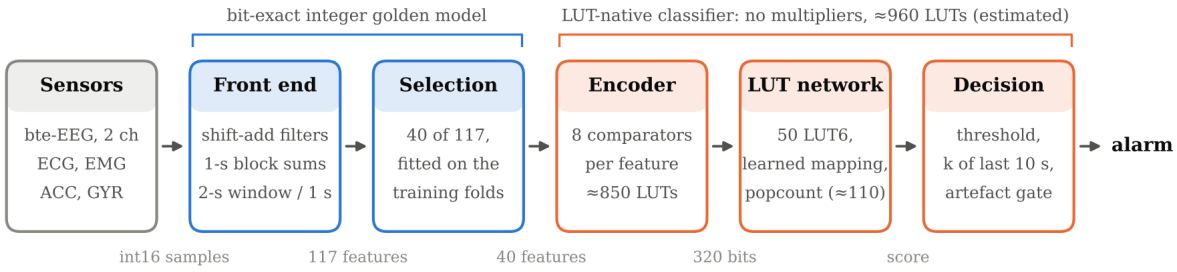


Figure 3.3 The proposed detector. Blue: bit-exact integer front end; orange: LUT-native classifier.

- Every second, the last 2 s of all signals become **117 integer features** (Table 3.4).
- The 40 most useful features are thermometer-encoded and classified by a **50-neuron LUT network**.
- An alarm is raised when at least k of the last 10 seconds are positive.

Table 3.4 The 117 integer features (per 2-s window; many also as change from a 64-s moving average).

Group	Count	Features
EEG	84	amplitude, line length, mobility, zero crossings, extrema, 5 band amplitudes; artefact flags
ECG	5	log heart period, log RR variability, quality flag
EMG	8	amplitude, line length, zero crossings, high/low band ratio
Motion	16	accelerometer and gyroscope magnitude, jerk, activity, band amplitude
Presence	4	one bit per sensor (ECG, EMG, accelerometer, gyroscope)

Integer arithmetic. Filters use coefficients that are sums of powers of two (shifts and adds only), with error feedback; heart rate uses $\log_2(\mathrm{RR})$, so there is no division. A floating-point twin measures the cost of the integer arithmetic (Section 4.2).

3.4 The LUT-Native Classifier (DWN)

A Differentiable Weightless Neural Network [9] has no weights: each neuron is a 6-input truth table, exactly one FPGA LUT.

$$b_{f,i} = [x_f > t_{f,i}], \quad i = 1, \ldots, 8 \qquad\qquad s = \sum_{j \in S} y_j - \sum_{j \in B} y_j \tag{3.1}$$

Each feature becomes 8 thermometer bits (one comparator each); each neuron reads 6 learned bits and outputs one stored bit y_j; the score s is the popcount of the seizure group S minus that of the background group B. Figure 3.4 shows the hardware cost of the final network.

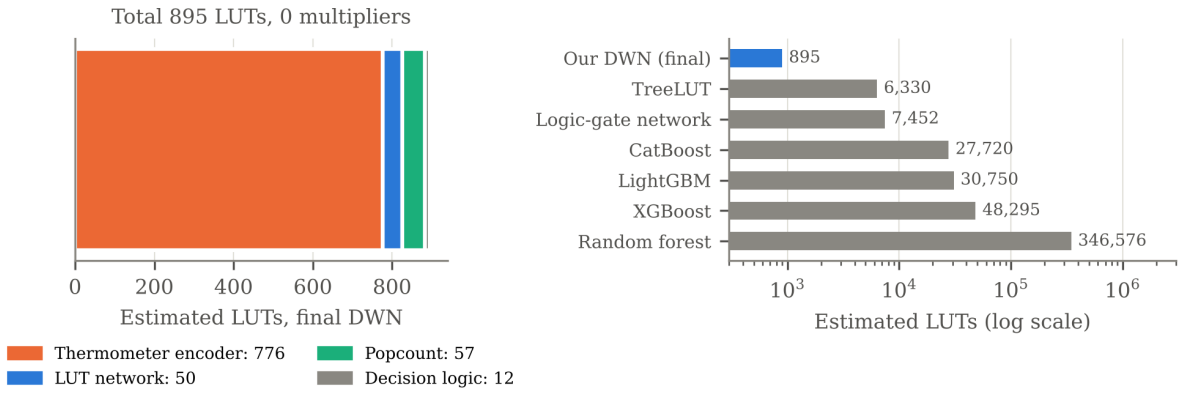


Figure 3.4 Estimated hardware cost. Left: the final DWN by block (the encoder dominates). Right: total LUTs of each model, log scale.

3.5 Models and Extensions

Table 3.5 Models compared on the same features, splits and metrics.

Group	Models
Published baselines	ChronoNet [28] (CNN + GRU on raw EEG); SVM [6]
Software references	logistic regression, random forest, XGBoost, LightGBM, CatBoost
Hardware-oriented	INT8 MLP (multipliers), TreeLUT [56], logic-gate network [10], DWN [9]

Table 3.6 Extensions of the DWN tested in this project (adopted only if significantly better).

ID	Idea
N1	Memory across seconds: 16 state bits fed back; stacked past windows
N2	Encoder design: uniform, Gaussian and learned thresholds
N3	Label-free per-patient threshold adaptation
N4	Distillation from LightGBM; teacher-guided hard negatives
N5	Focal, noise-robust (GCE, SCE) and multiple-instance losses; onset weighting
N6	Modality dropout, gated fusion, late fusion
N7	Interpretability: which inputs the LUTs read during alarms
N8	Post-ictal verification: a second LUT network confirms each alarm (secondary analysis)

3.6 Evaluation Protocol

- **Patient-wise 5-fold cross-validation** on the 96 training patients for every design choice.
- **Sealed test set**: the 29 test patients scored once, after the freeze, with 5 seeds; every access logged.
- **SzCORE event scoring** [40] (30 s before, 60 s after, 90 s merge), per patient:

$$\text{Sens} = \frac{TP}{TP+FN}, \qquad \text{Prec} = \frac{TP}{TP+FP}, \qquad F_1 = \frac{2\,\text{Sens}\cdot\text{Prec}}{\text{Sens}+\text{Prec}}, \qquad \text{false alarms per 24 h} \tag{3.2}$$

- **Operating points** (threshold, k, artefact gate) chosen on the training folds: best event-F1, and highest sensitivity with at most one false alarm per hour.
- **Statistics**: paired Wilcoxon test per patient; an extension is adopted only if $p < 0.05$ in its favour.

- **Paper metrics** (the base paper's own code) are reported for the comparison with the base paper.
- **Compute**: a 4-core CPU server and a laptop RTX 3060 GPU.

Chapter 4

Experimental Results and Comparisons

Sections 4.1–4.6: development on the 96 training patients (patient-wise cross-validation). Sections 4.7–4.10: the single evaluation on the 29 held-out test patients, after the design was frozen.

4.1 Reproduction of the Published Baseline

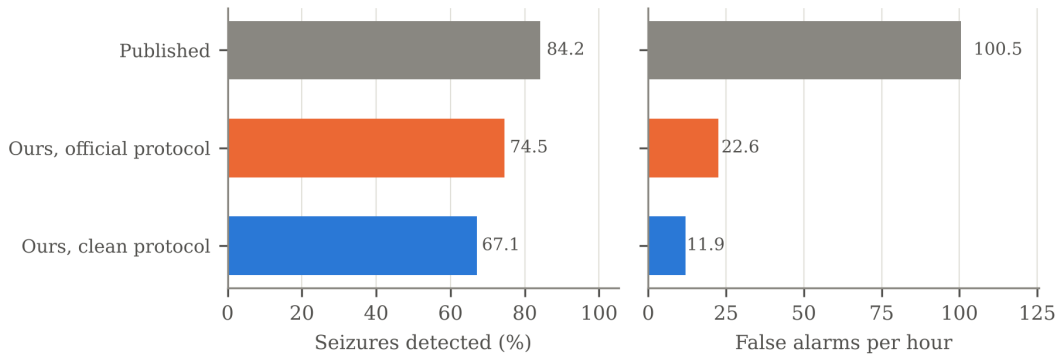


Figure 4.1 ChronoNet on the 29 test patients: published and reproduced (the paper's metrics).

- **Partly reproduced**: 74.5 % of seizures detected against 84.2 % published, with 22.6 instead of 100.5 false alarms per hour (Figure 4.1); the per-recording AUPR matches (0.109 vs 0.104).
- **Test-set leak**: the released code picks its training epoch on the test patients. Our **clean** run (last epoch, no test information: 67.1 %, 11.9 false alarms/h) is the reference for the final comparison.

4.2 Integer Versus Floating-Point Features

Table 4.1 The integer front end against its floating-point twin (10 training patients).

Check	Result
Median feature correlation, integer vs float	0.9965
Artefact flags agreeing	99.7 %
LightGBM AUROC, integer vs float features	0.736 vs 0.735

4.3 Comparison of Model Families

Table 4.2 Five-fold cross-validation on the 96 training patients (best-F1 operating point).

Model	AUROC	Event-F1	FA / 24 h	Sens. at $\leq$1 FA/h	Estimated cost
LightGBM	0.870	0.292	3.6	0.594	≈30.6k LUTs
CatBoost	0.863	0.286	3.0	0.618	≈27.7k LUTs
XGBoost	0.867	0.277	3.4	0.603	≈47.9k LUTs
TreeLUT	0.857	0.266	4.2	0.619	≈6.4k LUTs
DWN + distillation (final)	0.851	**0.229**	**0.9**	0.597	≈**870 LUTs**
DWN base	0.840	0.211	2.4	0.581	≈960 LUTs
Random forest	0.852	0.198	0.5	0.536	≈323k LUTs
Logic-gate network	0.843	0.171	4.9	0.540	≈7.5k LUTs
INT8 MLP	0.830	0.147	7.8	0.520	9,600 multiplications
Logistic regression	0.806	0.105	8.8	0.409	117 multiplications

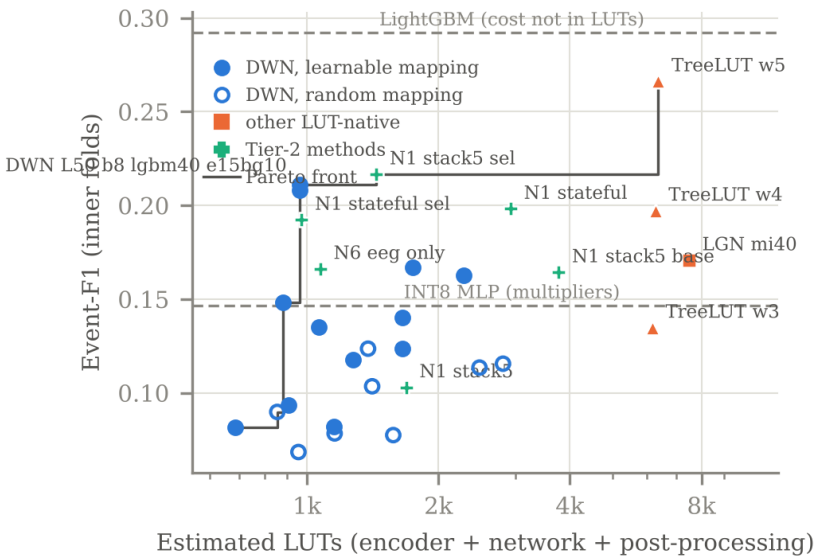


Figure 4.2 Event-F1 against estimated LUTs for every LUT-native run (training folds), with the Pareto front.

Findings. The DWN is 7–8× smaller than the other LUT models and beats the INT8 MLP on every metric (Table 4.2). Larger networks (up to 2,000 LUTs), 16 thresholds, two layers and other temperatures gave no significant gain (Figure 4.2); 4-input LUTs and 4 thresholds were clearly worse.

4.4 What Improved the Network

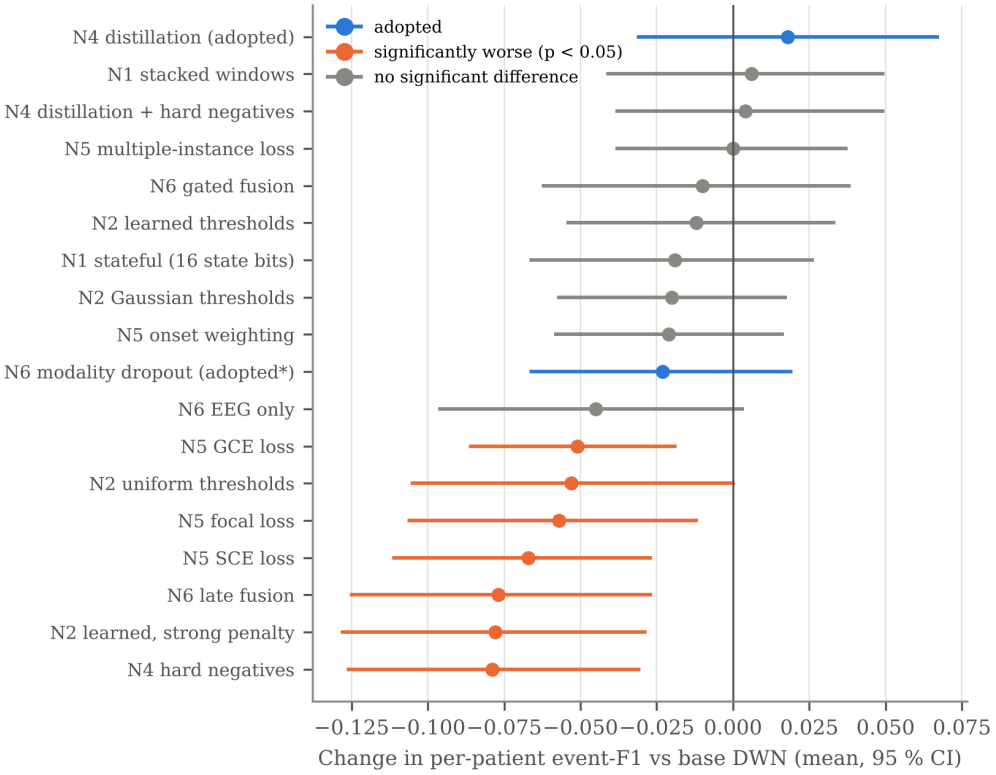


Figure 4.3 Every extension against the base DWN: mean change in per-patient event-F1 with 95 % interval. *Modality dropout was adopted for its robustness to missing sensors (Section 4.5).

Table 4.3 The changes that passed the significance test.

Change	Evidence (training patients)	How it was found
Feature selection by LightGBM importance	event-F1 0.148 → 0.208 ($p = 0.047$); false alarms 7.0 → 2.3/day	N7 showed the network ignored heart rate
Distillation from LightGBM (**final**)	false alarms 2.4 → 0.9/day ($p = 3 \times 10^{-7}$); event-F1 0.229 vs 0.211	N4
Modality dropout (variant)	without ECG: event-F1 0.092 → 0.172 ($p = 0.005$)	N6

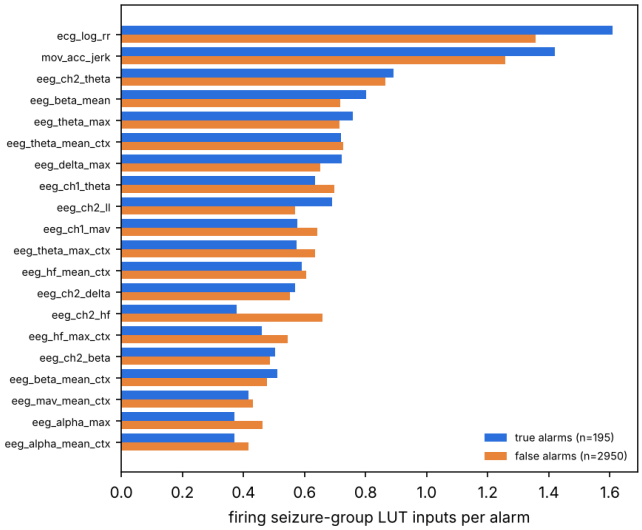


Figure 4.4 N7: inputs read by the firing seizure LUTs during true and false alarms (first, mutual-information network): false alarms look like seizures to it.

4.5 Missing Sensors

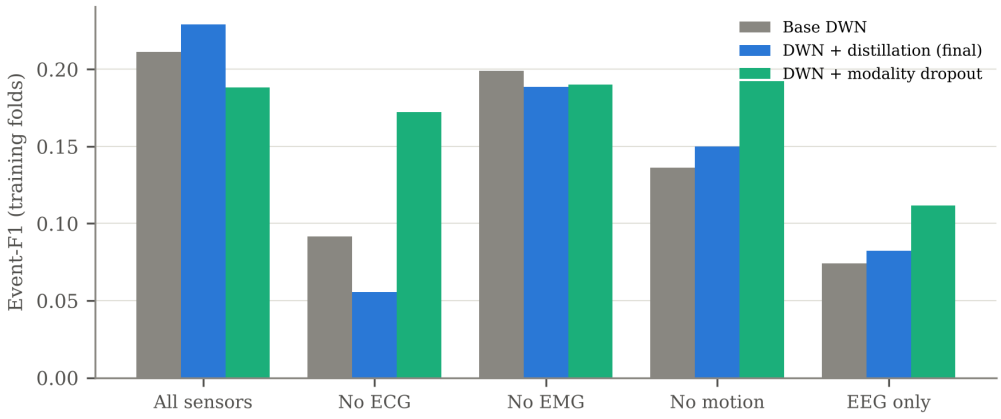


Figure 4.5 Event-F1 with sensors removed at test time (training folds). Distillation is the most accurate with all sensors; modality dropout keeps working without ECG or motion.

4.6 Seizure Types

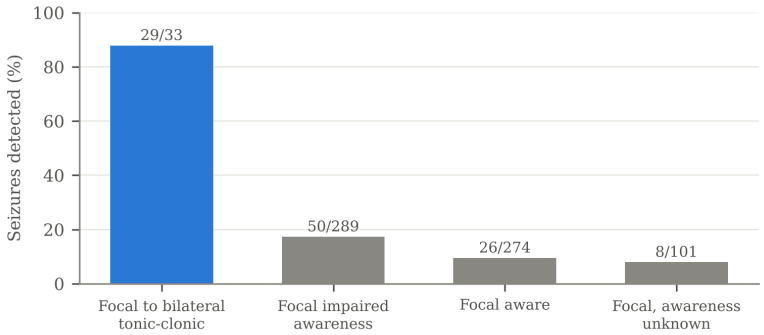


Figure 4.6 Seizures detected by type (base DWN, training folds): tonic-clonic seizures are caught, focal seizures rarely.

4.7 Final Evaluation on the Held-Out Test Patients

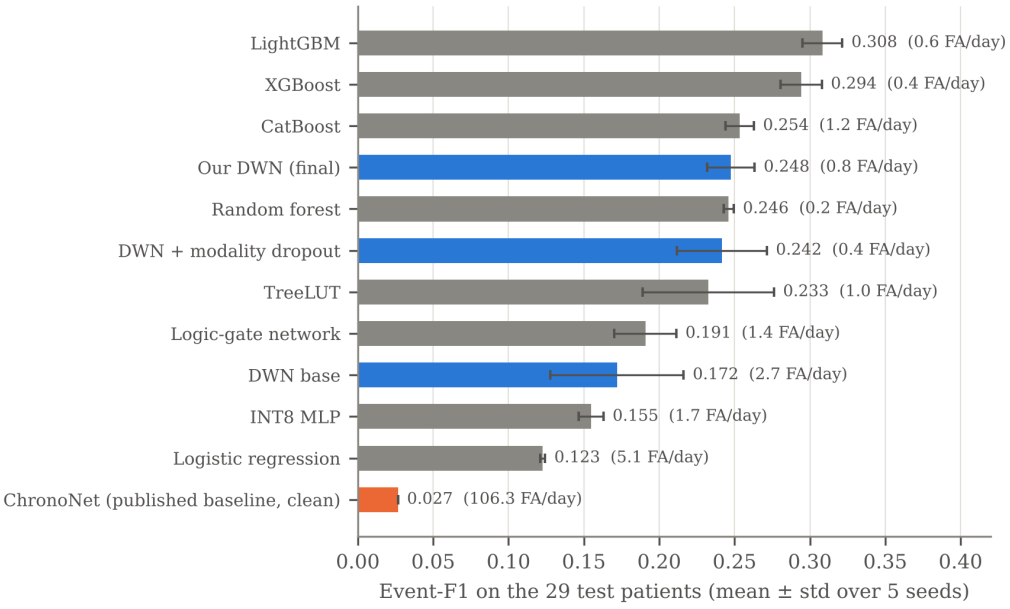


Figure 4.7 Event-F1 on the 29 test patients (mean ± std over 5 seeds) and false alarms per day.

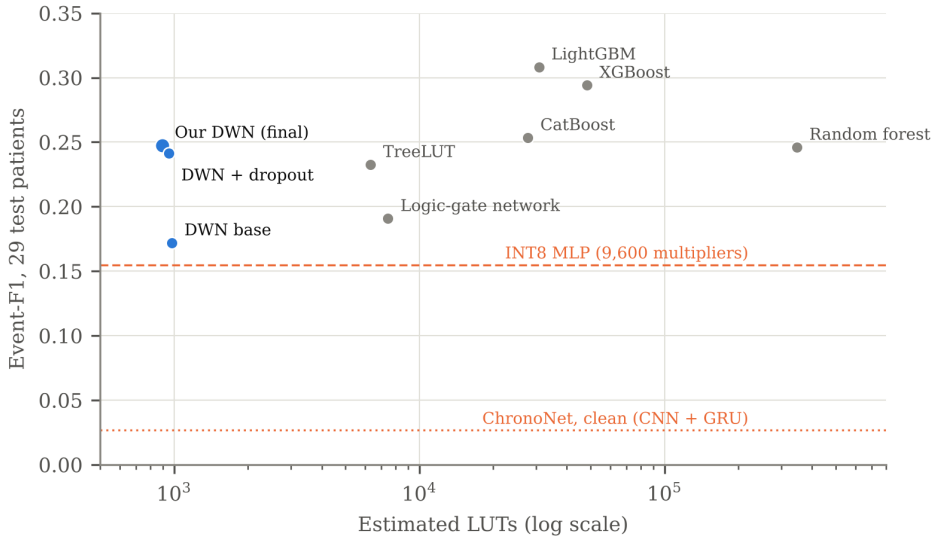


Figure 4.8 Accuracy against hardware cost on the test patients: the LUT networks reach tree-ensemble accuracy at a fraction of the LUTs.

Table 4.4 Final DWN against the main references on the 29 test patients (p: paired Wilcoxon on per-patient event-F1).

Model	Event-F1	FA / 24 h	p vs ChronoNet	Estimated cost
DWN + distillation (final)	**0.248 ± 0.016**	**0.8**	0.027	**≈900 LUTs**
LightGBM	0.308 ± 0.013	0.6	0.027	≈30.8k LUTs
CatBoost	0.254 ± 0.009	1.2	0.031	≈27.7k LUTs
Random forest	0.246 ± 0.003	0.2	0.27	≈347k LUTs
TreeLUT	0.233 ± 0.044	1.0	0.016	≈6.3k LUTs
Logic-gate network	0.191 ± 0.021	1.4	0.034	≈7.5k LUTs
INT8 MLP	0.155 ± 0.008	1.7	0.017	9,600 multiplications
ChronoNet (clean)	0.027	106	–	CNN + GRU

Findings. The result held on unseen patients (0.248, above the cross-validation 0.229). The final DWN scores above TreeLUT, the logic-gate network and the INT8 MLP, is level with random forest and CatBoost at 30–390× fewer LUTs, and is significantly better than ChronoNet; LightGBM remains ahead ($p = 0.039$). All 11 models are in Table A.1 (Appendix A).

4.8 Comparison with the Base Paper

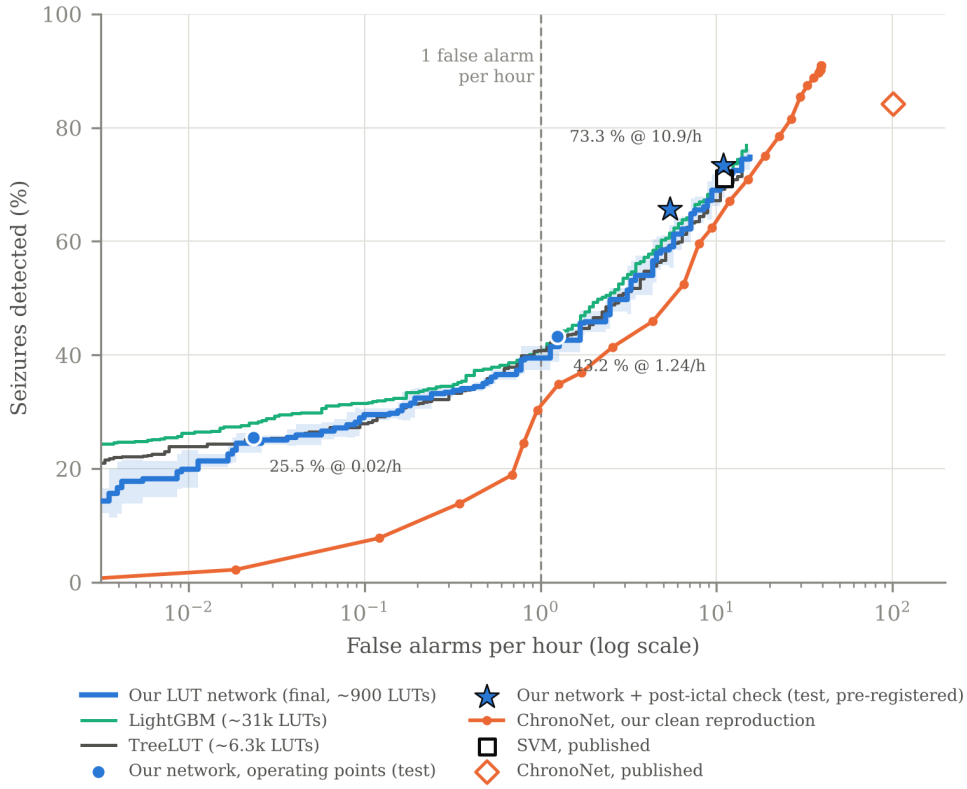


Figure 4.9 Seizures detected against false alarms per hour on the 29 test patients (the base paper's metrics). Curves: threshold sweeps of the frozen models (mean of 5 seeds; band: std of our network), drawn as far as every seed's sweep reaches; dots: our two operating points, chosen on the training folds; stars: the two-stage detector (Section 4.9); open markers: published values.

Table 4.5 The base paper's baselines and our network on the same 29 test patients (the paper's metrics).

Model	Detected	FA / h	AUROC
SVM, published [8]	71.1 %	11.0	0.700
ChronoNet, published [8]	84.2 %	100.5	0.792
ChronoNet, our clean reproduction	67.1 %	11.9	0.746
ChronoNet at ≈1.3 FA/h*	34.8 %	1.26	–
Our DWN, ≤1 FA/h operating point	**43.2 %**	**1.24**	**0.883**
Our DWN, best-F1 operating point	**25.5 %**	**0.02**	
Our DWN + post-ictal check (pre-registered)†	**73.3 %**	**10.9**	
Same, stricter verification†	**65.6 %**	**5.4**	

* threshold read from ChronoNet's own test curve (favours it); at 0.02 FA/h ChronoNet detects 2.2 %. AUROC pooled over test windows; the published 0.792 uses an unreproduced definition. † Secondary analysis after the final test (Section 4.9); 96 s decision delay.

Findings. The paper's baselines run at 11–100 false alarms per hour. At the same false-alarm rate our network detects more seizures than ChronoNet (43.2 % vs 34.8 % at ≈1.25/h; 25.5 % vs 2.2 % at ≈0.02/h), ranks windows better (AUROC 0.883 vs 0.746), and needs no multipliers. At the SVM's rate of 11 false alarms per hour its curve reaches 69.7 % (SVM, published: 71.1 %), level with LightGBM (70.5 %) at 3 % of its LUTs; this value is read from the test curve itself and is therefore optimistic.

4.9 Post-Ictal Verification (Secondary Analysis)

Requested after the final test: detect more seizures than the base paper at slightly fewer false alarms. A two-stage detector (N8) runs the final network at a sensitive threshold and confirms each alarm 96 s later with a second LUT network (50 LUTs) that compares the alarm with the minutes before and after it (EEG, heart rate, EMG, motion). All settings were chosen on the training patients and committed before one evaluation on the test patients.

Table 4.6 Two-stage detector: training patients (cross-validation) and test patients (5 seeds, paper's metrics).

Detector	**Training (CV)**		**Test**	
	Detected	FA/h	Detected	FA/h
Final network, lower threshold	74.2 %	8.5	66.4 ± 2.0 %	7.6
+ post-ictal check (≤10 FA/h)	81.9 %	9.6	**73.3 ± 1.4 %**	**10.9**
+ post-ictal check (≤5 FA/h)	72.3 %	4.8	65.6 ± 1.0 %	5.4
SVM, published [8]	–	–	71.1 %	11.0

Findings. At the SVM's false-alarm rate (10.9 vs 11.0 per hour) the LUT-native two-stage detector detects 2.2 points more seizures (all 5 seeds above 71.1 %; a small margin against a single published value). Verification adds 3.6–6.5 points over the network alone at the same false-alarm rate. Costs: a 96 s decision delay (seizure logging, not instant alarms) and many false alarms per day, as for the paper's baselines; the low-false-alarm result of Section 4.7 remains the primary one.

4.10 Hardware Handoff Check

Table 4.7 Verilog of the final DWN against the software on 10 test recordings.

Item	**Result**
Exported	thresholds, input mapping, 50 truth tables, popcount head
Test vectors	raw samples, features, bits, LUT outputs, scores, alarms
Windows simulated (Icarus Verilog)	431,960
Mismatches	**0**

Chapter 5

Challenges Faced and How They Were Addressed

Table 5.1 The main problems of Phase I and their solutions.

Area	Challenge	What happened	Solution
Data	Files differ from the documentation	256 Hz not 250 Hz; 883 not 886 seizures; no centre or age labels	Full audit first; design follows the files
Data	126 GB, no local disk or GPU	Dataset too large for the server	Stream, extract, delete: only 3.5 GB of features kept
Data	Silent download truncation	9 recordings cut short (one EEG file: 12 of 16,363 windows)	Size checks; re-extracted; 3 runs repeated
Data	Official artefact rule	Flags 69 % of normal behind-the-ear windows	Flags used as input bits; gating chosen on training folds
Baseline	ChronoNet only partly reproduced	74.5 % vs 84.2 % detected	Literal port (identical on 900 test cases); gap reported as measured
Baseline	Released code tunes on the test patients	Best epoch picked on the evaluation patients	Clean run with no test information as our reference
Baseline	SVM features unavailable	Defined in a paper we could not access	Open; requested for Phase II
Hardware	Rounding noise in integer filters	Error 140 LSB, larger than the signal (61 LSB)	Error feedback: 0.7 LSB

Area	Challenge	What happened	Solution
Hardware	Heart rate needs a division	$60/\mathrm{RR}$	$\log_2(\mathrm{RR})$: no division or multiplication
Hardware	Verilog disagreed with software	Thresholds beyond a feature's bit width	Generator fixed: 0 mismatches
Compute	6 GB laptop GPU; CUDA-only library	DWN could not run on the CPU server	PyTorch re-implementation, unit-tested against the kernel
Compute	Interrupted runs; out of memory	Lost folds; a sampler used 13.9 GB	Resumable runs; new sampler peaks at 3.1 GB
ML	Seizures are 0.123 % of the time	Accuracy is meaningless	Event scoring per patient; balanced sampling; false-alarm budget
ML	**Weak early results**	First network: event-F1 0.148, 7.0 false alarms/day	N7 showed heart rate was ignored; new feature selection: 0.208, 2.3/day
ML	Bugs in comparisons	Wrong feature set; distillation lost on retraining	Found in run records; shared training code, tests, runs repeated

Chapter 6

Discussion, Conclusions and Future Scope

6.1 Key Findings

- **Accuracy per LUT**: the 50-neuron network (about 900 LUTs, no multipliers) matches random forest and CatBoost at 30–390× fewer LUTs and scores above every other hardware model; LightGBM is still more accurate.
- **Far fewer false alarms than the base paper**: 0.02 per hour (paper's metric) against 11–100 per hour for the published baselines; at an equal false-alarm rate it detects more seizures than ChronoNet (Figure 4.9).
- **Above the base paper's SVM at its false-alarm rate**: with post-ictal verification, 73.3 % of seizures at 10.9 false alarms per hour against 71.1 % at 11.0 (pre-registered, test patients; Section 4.9).
- **The encoder is the cost**: 776 of the 895 LUTs are thermometer comparators; the network itself is 50 (Figure 3.4).
- **Looking inside the LUTs paid off**: N7 found the ignored heart signal, the largest single improvement.
- **Focal seizures remain hard**: tonic-clonic seizures are detected, focal seizures rarely (Figure 4.6), as in the literature [4].
- **Honest evaluation matters**: the published baseline tunes on its test patients; ours were sealed until the design freeze.

6.2 Status at the End of Phase I

Table 6.1 Status of the project activities.

Activity	Status	Result
Literature review	Done	79 references; gaps identified
Dataset audit	Done	11,009 files measured
ChronoNet reproduction (official and clean)	Done	74.5 % / 67.1 % detected
SVM reproduction	Pending	needs the original feature list
Integer feature pipeline	Done	117 features; no accuracy loss
Model families and DWN design study	Done	9 families; DWN of 50–2,000 LUTs
Extensions N1–N7	Done	distillation and modality dropout adopted
Final test (29 patients, 5 seeds)	Done	event-F1 0.248; 0.8 false alarms/day
Verilog handoff	Done	0 mismatches on 431,960 windows
FPGA implementation	Phase II	synthesis, power, latency

Table 6.2 Work carried out during Phase I.

Period	Activities
Month 1 (September 2026)	Topic and problem definition with the guide; literature study; study of SeizeIT2, its baselines and metrics; tools and data access; first experiments (weaker results)
Month 2 (October – November 2026)	Dataset audit; ChronoNet reproduction; integer pipeline; model comparison; DWN design study and extensions; design freeze; final test; Verilog check; this report

6.3 Plan for Phase II

Table 6.3 Month-wise plan for Phase II.

Month	Work	Deliverable
December 2026	FPGA tool flow; synthesis and place-and-route of the verified design	Resource and timing report
January 2027	Conference paper (EMBC 2027, full papers reportedly due 24 January 2027)	Submitted paper
February 2027	Measured LUTs, power and latency; INT8 MLP and TreeLUT on the same FPGA	Hardware comparison
March 2027	Recordings streamed through the board; focal-seizure study on training patients	Demonstration
April 2027	Final report and presentation	Final report

6.4 Conclusions

1. SeizeIT2 was audited; the design follows the files, not the documentation.
2. ChronoNet was partly reproduced, and a clean version (no test information) is our reference.
3. The integer multimodal front end loses no measurable accuracy against floating point.
4. A LUT-native network with distillation reaches an event-F1 of 0.248 with 0.8 false alarms per day on 29 unseen patients at about 900 LUTs and no multipliers: significantly better than ChronoNet, level with random forest and CatBoost, below LightGBM.
5. Its Verilog matches the software exactly; Phase II takes it to an FPGA.

References

[1] GBD 2016 Epilepsy Collaborators, "Global, regional, and national burden of epilepsy, 1990–2016: a systematic analysis for the global burden of disease study 2016," *The Lancet Neurology*, vol. 18, no. 4, pp. 357–375, 2019.

[2] C. E. Elger and C. Hoppe, "Diagnostic challenges in epilepsy: seizure under-reporting and seizure detection," *The Lancet Neurology*, vol. 17, no. 3, pp. 279–288, 2018.

[3] P. Ryvlin, L. Nashef, S. D. Lhatoo *et al.*, "Incidence and mechanisms of cardiorespiratory arrests in epilepsy monitoring units (MORTEMUS): a retrospective study," *The Lancet Neurology*, vol. 12, no. 10, pp. 966–977, 2013.

[4] S. Beniczky, S. Wiebe, J. Jeppesen, W. O. Tatum, M. Brázdil, Y. Wang, S. T. Herman, and P. Ryvlin, "Automated seizure detection using wearable devices: A clinical practice guideline of the International League Against Epilepsy and the International Federation of Clinical Neurophysiology," *Epilepsia*, vol. 62, no. 3, pp. 632–646, 2021.

[5] L. Swinnen *et al.*, "A multicenter, video-EEG-based validation of a multimodal wearable device for focal seizure detection in adults: The SeizeIT2 study," *Epilepsia Open*, 2026.

[6] K. Vandecasteele, T. De Cooman, J. Dan, E. Cleeren, S. Van Huffel, B. Hunyadi, and W. Van Paesschen, "Visual seizure annotation and automated seizure detection using behind-the-ear electroencephalographic channels," *Epilepsia*, vol. 61, no. 4, pp. 766–775, 2020.

[7] K. Vandecasteele, T. De Cooman, C. Chatzichristos, E. Cleeren, L. Swinnen, J. Macea Ortiz, S. Van Huffel, M. Dümpelmann, A. Schulze-Bonhage, M. De Vos, W. Van Paesschen, and B. Hunyadi, "The power of ECG in multimodal patient-specific seizure monitoring," *Epilepsia*, vol. 62, no. 10, pp. 2333–2343, 2021.

[8] M. Bhagubai, C. Chatzichristos, L. Swinnen, J. Macea, J. Zhang, L. Lagae, K. Jansen, A. Schulze-Bonhage, F. Sales, B. Mahler, Y. Weber, W. Van Paesschen, and M. De Vos, "SeizeIT2: Wearable dataset of patients with focal epilepsy," *Scientific Data*, vol. 12, p. 1228, 2025.

[9] A. T. L. Bacellar, Z. Susskind, M. Breternitz Jr., E. John, L. K. John, P. M. V. Lima, and F. M. G. França, "Differentiable weightless neural networks," in *Proceedings of the 41st International Conference on Machine Learning (ICML)*, ser. Proceedings of Machine Learning Research, vol. 235. PMLR, 2024, pp. 2277–2295. [Online]. Available: https://proceedings.mlr.press/v235/bacellar24a.html

[10] F. Petersen, C. Borgelt, H. Kuehne, and O. Deussen, "Deep differentiable logic gate networks," in *Advances in Neural Information Processing Systems (NeurIPS)*, vol. 35, 2022. [Online]. Available: https://openreview.net/forum?id=vF3WefcoePW

[11] Y. Umuroglu, Y. Akhauri, N. J. Fraser, and M. Blott, "LogicNets: Co-designed neural networks and circuits for extreme-throughput applications," in *30th International Conference on Field-Programmable Logic and Applications (FPL)*, 2020, pp. 291–297.

[12] R. S. Fisher, J. H. Cross, J. A. French, N. Higurashi, E. Hirsch, F. E. Jansen, L. Lagae, S. L. Moshé, J. Peltola, E. Roulet Perez, I. E. Scheffer, and S. M. Zuberi, "Operational classification of seizure types by the International League Against Epilepsy: Position paper of the ILAE commission for classification and terminology," *Epilepsia*, vol. 58, no. 4, pp. 522–530, 2017.

[13] F. S. S. Leijten and Dutch TeleEpilepsy Consortium, "Multimodal seizure detection: A review," *Epilepsia*, vol. 59, no. S1, pp. 42–47, 2018.

[14] S. Beniczky, I. Conradsen, O. Henning, M. Fabricius, and P. Wolf, "Automated real-time detection of tonic-clonic seizures using a wearable EMG device," *Neurology*, vol. 90, no. 5, pp. e428–e434, 2018.

[15] F. Onorati, G. Regalia, C. Caborni, M. Migliorini, D. Bender, M.-Z. Poh, C. Frazier, E. Kovitch Thropp, E. D. Mynatt, J. Bidwell, R. Mai, W. C. LaFrance, Jr., A. S. Blum, D. Friedman, T. Loddenkemper, F. Mohammadpour-Touserkani, C. Reinsberger, S. Tognetti, and R. W. Picard, "Multicenter clinical assessment of improved wearable multimodal convulsive seizure detectors," *Epilepsia*, vol. 58, no. 11, pp. 1870–1879, 2017.

[16] G. Regalia, F. Onorati, M. Lai, C. Caborni, and R. W. Picard, "Multimodal wrist-worn devices for seizure detection and advancing research: Focus on the Empatica wristbands," *Epilepsy Research*, vol. 153, pp. 79–82, 2019.

[17] V. Naganur, S. Sivathamboo, Z. Chen, S. Kusmakar, A. Antonic-Baker, T. J. O'Brien, and P. Kwan, "Automated seizure detection with noninvasive wearable devices: A systematic review and meta-analysis," *Epilepsia*, vol. 63, pp. 1930–1941, 2022.

[18] M. Bhagubai, K. Vandecasteele, L. Swinnen, J. Macea, C. Chatzichristos, M. De Vos, and W. Van Paesschen, "The power of ECG in semi-automated seizure detection in addition to two-channel behind-the-ear EEG," *Bioengineering*, vol. 10, no. 4, p. 491, 2023.

[19] M. Bhagubai, L. Swinnen, E. Cleeren, W. Van Paesschen, M. De Vos, and C. Chatzichristos, "Towards automated seizure detection with wearable EEG – grand challenge," *IEEE Open Journal of Signal Processing*, vol. 5, pp. 717–724, 2024.

[20] C. Reintjes, J. F. Hagenbeck, M. Ballo, T. Rahlmeier, S. M. Wolf, and D. Schoder, "ECG-based detection of epileptic seizures in real-world wearable settings: Insights from the SeizeIT2 dataset," *Sensors*, vol. 25, no. 24, p. 7687, 2025.

[21] T. Su, H. Zhu, S. Chen, and H. Wang, "CMEpiNet: Complex-valued multimodal epilepsy detection network model," *Sensors*, 2026.

[22] J. Zhang, L. Swinnen, C. Chatzichristos, V. Broux, R. Proost, K. Jansen, B. Mahler, N. Zabler, N. Epitashvilli, M. Dümpelmann, A. Schulze-Bonhage, E. Schriewer, U. Ermis, S. Wolking, F. Linke, Y. Weber, M. Symmonds, A. Sen, A. Biondi, M. P. Richardson, A. Sulaiman I, A. I. Silva, F. Sales, G. Vértes, W. Van Paesschen, and M. De Vos, "Multimodal wearable EEG, EMG and accelerometry measurements improve the accuracy of tonic-clonic seizure detection," *Physiological Measurement*, vol. 45, no. 6, 2024.

[23] J. Zhang, L. Swinnen, C. Chatzichristos, W. Van Paesschen, and M. De Vos, "Learning robust representations of tonic-clonic seizures with cyclic transformer," *IEEE Journal of Biomedical and Health Informatics*, vol. 28, no. 6, pp. 3721–3731, 2024.

[24] T. M. Ingolfsson, X. Wang, U. Chakraborty, S. Benatti, A. Bernini, P. Ducouret, P. Ryvlin, S. Beniczky, L. Benini, and A. Cossettini, "BrainFuseNet: Enhancing wearable seizure detection through EEG-PPG-accelerometer sensor fusion and efficient edge deployment," *IEEE Transactions on Biomedical Circuits and Systems*, vol. 18, no. 4, pp. 720–733, 2024.

[25] A. Shoeb and J. Guttag, "Application of machine learning to epileptic seizure detection," in *Proceedings of the 27th International Conference on Machine Learning (ICML)*, 2010, pp. 975–982. [Online]. Available: https://archive.physionet.org/pn6/chbmit/shoeb-icml-2010.pdf

[26] I. Al-Hussaini and C. S. Mitchell, "SeizFt: Interpretable machine learning for seizure detection using wearables," *Bioengineering*, vol. 10, no. 8, p. 918, 2023.

[27] B. Huang, A. Abtahi, and A. Aminifar, "Lightweight machine learning for seizure detection on wearable devices," in *IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*, 2023. [Online]. Available: https://portal.research.lu.se/en/publications/lightweight-machine-learning-for-seizure-detection-on-wearable-de/

[28] S. Roy, I. Kiral-Kornek, and S. Harrer, "ChronoNet: A deep recurrent neural network for abnormal EEG identification," in *Artificial Intelligence in Medicine (AIME 2019)*, ser. Lecture Notes in Computer Science, vol. 11526. Springer, 2019.

[29] V. J. Lawhern, A. J. Solon, N. R. Waytowich, S. M. Gordon, C. P. Hung, and B. J. Lance, "EEGNet: a compact convolutional neural network for EEG-based brain–computer interfaces," *Journal of Neural Engineering*, vol. 15, no. 5, p. 056013, 2018.

[30] P. Thuwajit, P. Rangpong, P. Sawangjai, P. Autthasan, R. Chaisaen, N. Banluesombatkul, P. Boonchit, N. Tatsaringkansakul, T. Sudhawiyangkul, and T. Wilaiprasitporn, "EEGWaveNet: Multiscale CNN-based spatiotemporal feature extraction for EEG seizure detection," *IEEE Transactions on Industrial Informatics*, vol. 18, no. 8, pp. 5547–5557, 2022.

[31] S. Tang, J. Dunnmon, K. K. Saab, X. Zhang, Q. Huang, F. Dubost, D. Rubin, and C. Lee-Messer, "Self-supervised graph neural networks for improved electroencephalographic seizure analysis," in *International Conference on Learning Representations (ICLR)*, 2022. [Online]. Available: https://openreview.net/forum?id=k9bx1EfHI_-

[32] K. Wu, Z. Zhao, and B. Yener, "SeizureTransformer: Scaling U-Net with transformer for simultaneous time-step level seizure detection from long EEG recordings," 2025.

[33] C. Yang, M. B. Westover, and J. Sun, "BIOT: Biosignal transformer for cross-data learning in the wild," in *Advances in Neural Information Processing Systems (NeurIPS)*, vol. 36, 2023.

[34] K. Saab, J. Dunnmon, C. Ré, D. Rubin, and C. Lee-Messer, "Weak supervision as an efficient approach for automated seizure detection in electroencephalography," *npj Digital Medicine*, vol. 3, 2020.

[35] D. Y. Isaev, D. Tchapyjnikov, C. M. Cotten, D. Tanaka, N. Martinez, M. Bertran, G. Sapiro, and D. Carlson, "Attention-based network for weak labels in neonatal seizure detection," in *Proceedings of the 5th Machine Learning for Healthcare Conference*, ser. Proceedings of Machine Learning Research, vol. 126, 2020, pp. 479–507. [Online]. Available: https://proceedings.mlr.press/v126/isaev20a.html

[36] D. M. Shama and A. Venkataraman, "Bayesian uncertainty-aware deep learning with noisy labels: Tackling annotation ambiguity in EEG seizure detection," *PLOS ONE*, vol. 21, no. 6, 2026.

[37] Q. Zheng, A. Venkitaraman, S. Petravic, and P. Frossard, "Knowledge distillation with graph neural networks for epileptic seizure detection," in *Machine Learning and Knowledge Discovery in Databases (ECML PKDD 2023)*, ser. Lecture Notes in Computer Science. Springer, 2023, pp. 547–563.

[38] S. Ziyabari, V. Shah, M. Golmohammadi, I. Obeid, and J. Picone, "Objective evaluation metrics for automatic classification of EEG events," 2017.

[39] V. Shah, E. von Weltin, S. Lopez, J. R. McHugh, L. Veloso, M. Golmohammadi, I. Obeid, and J. Picone, "The Temple University Hospital seizure detection corpus," *Frontiers in Neuroinformatics*, 2018.

[40] J. Dan, U. Pale, A. Amirshahi, W. Cappelletti, T. M. Ingolfsson, X. Wang, A. Cossettini, A. Bernini, L. Benini, S. Beniczky, D. Atienza, and P. Ryvlin, "SzCORE: Seizure community open-source research evaluation framework for the validation of electroencephalography-based automated seizure detection algorithms," *Epilepsia*, 2024.

[41] J. Dan, A. Shahbazinia, C. Kechris, and D. Atienza, "SzCORE as a benchmark: report from the seizure detection challenge at the 2025 AI in epilepsy and neurological disorders conference," 2025.

[42] G. Brookshire, J. Kasper, N. M. Blauch, Y. C. Wu, R. Glatt, D. A. Merrill, S. Gerrol, K. J. Yoder, C. Quirk, and C. Lucero, "Data leakage in deep learning studies of translational EEG," *Frontiers in Neuroscience*, vol. 18, p. 1373515, 2024.

[43] I. Aleksander, W. V. Thomas, and P. A. Bowden, "WISARD·a radical step forward in image recognition," *Sensor Review*, vol. 4, no. 3, pp. 120–124, 1984.

[44] Z. Susskind, A. Arora, I. D. S. Miranda, L. A. Q. Villon, R. F. Katopodis, L. S. de Araújo, D. L. C. Dutra, P. M. V. Lima, F. M. G. França, M. Breternitz Jr., and L. K. John, "Weightless neural networks for efficient edge inference," in *Proceedings of the International Conference on Parallel Architectures and Compilation Techniques (PACT '22)*. ACM, 2022, pp. 279–290.

[45] Z. Susskind, A. Arora, I. D. S. Miranda, A. T. L. Bacellar, L. A. Q. Villon, R. F. Katopodis, L. S. de Araújo, D. L. C. Dutra, P. M. V. Lima, F. M. G. França, M. Breternitz Jr., and L. K. John, "ULEEN: A novel architecture for ultra low-energy edge neural networks," *ACM Transactions on Architecture and Code Optimization*, vol. 20, no. 4, p. 61, 2023.

[46] A. T. L. Bacellar, M. P. Jadhao, S. Nag, P. M. V. Lima, F. M. G. França, and L. K. John, "nanoML for human activity recognition," in *tinyML Research Symposium*, 2025.

[47] F. Kresse and C. H. Lampert, "Differentiable weightless controllers: Learning logic circuits for continuous control," 2025.

[48] F. Petersen, H. Kuehne, C. Borgelt, J. Welzel, and S. Ermon, "Convolutional differentiable logic gate networks," in *Advances in Neural Information Processing Systems (NeurIPS)*, vol. 37, 2024.

[49] L. Rüttgers, T. Aczel, A. Plesner, and R. Wattenhofer, "Light differentiable logic gate networks," in *International Conference on Learning Representations (ICLR)*, 2026.

[50] S. Bührer, A. Plesner, T. Aczel, and R. Wattenhofer, "Recurrent deep differentiable logic gate networks," 2025.

[51] E. Wang, J. J. Davis, P. Y. K. Cheung, and G. A. Constantinides, "LUTNet: Rethinking inference in FPGA soft logic," in *IEEE 27th Annual International Symposium on Field-Programmable Custom Computing Machines (FCCM)*, 2019, pp. 26–34.

[52] M. Andronic and G. A. Constantinides, "PolyLUT: Learning piecewise polynomials for ultra-low latency FPGA LUT-based inference," in *International Conference on Field Programmable Technology (ICFPT)*, 2023, pp. 60–68.

[53] ——, "NeuraLUT: Hiding neural network density in boolean synthesizable functions," in *34th International Conference on Field-Programmable Logic and Applications (FPL)*, 2024, pp. 140–148.

[54] O. Weng, M. Andronic, D. Zuberi, J. Chen, C. Geniesse, G. A. Constantinides, N. Tran, N. J. Fraser, J. M. Duarte, and R. Kastner, "Greater than the sum of its LUTs: Scaling up LUT-based neural networks with AmigoLUT," in *Proceedings of the 2025 ACM/SIGDA International Symposium on Field Programmable Gate Arrays (FPGA '25)*. ACM, 2025, pp. 25–35.

[55] O. Cassidy, M. Andronic, S. Coward, and G. A. Constantinides, "ReducedLUT: Table decomposition with "don't care" conditions," in *Proceedings of the 2025 ACM/SIGDA International Symposium on Field Programmable Gate Arrays (FPGA '25)*. ACM, 2025, pp. 36–42.

[56] A. Khataei and K. Bazargan, "TreeLUT: An efficient alternative to deep neural networks for inference acceleration using gradient boosted decision trees," in *Proceedings of the 2025 ACM/SIGDA International Symposium on Field Programmable Gate Arrays (FPGA '25)*. ACM, 2025.

[57] M. Mecik and M. Kumm, "Implementation and analysis of thermometer encoding in DWN FPGA accelerators," in *Asilomar Conference on Signals, Systems, and Computers*, 2025.

[58] W. Mommen, L. Keuninckx, P. Detterer, A. Colpaert, and P. Wambacq, "Inter-patient ECG arrhythmia classification with LGNs and LUTNs," 2026.

[59] S. Dharia, S. Smith, and C. Valderrama, "Differentiable logic gate networks for low-latency EEG classification on edge devices," in *Proceedings of the 39th Canadian Conference on Artificial Intelligence*, ser. Proceedings of Machine Learning Research, vol. 318. PMLR, 2026, pp. 377–391. [Online]. Available: https://proceedings.mlr.press/v318/dharia26a.html

[60] K. de Aguiar, F. M. G. França, V. C. Barbosa, and C. A. D. Teixeira, "Early detection of epilepsy seizures based on a weightless neural network," in *37th Annual International Conference of the IEEE Engineering in Medicine and Biology Society (EMBC)*, 2015, pp. 4470–4474.

[61] N. Verma, A. Shoeb, J. Bohorquez, J. Dawson, J. Guttag, and A. P. Chandrakasan, "A micro-power EEG acquisition SoC with integrated feature extraction processor for a chronic seizure detection system," *IEEE Journal of Solid-State Circuits*, vol. 45, no. 4, pp. 804–816, 2010.

[62] M. A. Bin Altaf, C. Zhang, and J. Yoo, "A 16-channel patient-specific seizure onset and termination detection SoC with impedance-adaptive transcranial electrical stimulator," *IEEE Journal of Solid-State Circuits*, vol. 50, no. 11, pp. 2728–2740, 2015.

[63] M. A. Bin Altaf and J. Yoo, "A 1.83 μJ/classification, 8-channel, patient-specific epileptic seizure classification SoC using a non-linear support vector machine," *IEEE Transactions on Biomedical Circuits and Systems*, vol. 10, 2016.

[64] S.-A. Huang, K.-C. Chang, H.-H. Liou, and C.-H. Yang, "A 1.9-mW SVM processor with on-chip active learning for epileptic seizure control," *IEEE Journal of Solid-State Circuits*, vol. 55, no. 2, pp. 452–464, 2020.

[65] M. Shoaran, B. Allahgholizadeh Haghi, M. Taghavi, M. Farivar, and A. Emami-Neyestanak, "Energy-efficient classification for resource-constrained biomedical applications," *IEEE Journal on Emerging and Selected Topics in Circuits and Systems*, vol. 8, no. 4, pp. 693–707, 2018.

[66] U. Shin, C. Ding, B. Zhu, Y. Vyza, A. Trouillet, E. Revol, S. Lacour, and M. Shoaran, "NeuralTree: A 256-channel 0.227-μJ/class versatile neural activity classification and closed-loop neuromodulation SoC," *IEEE Journal of Solid-State Circuits*, vol. 57, no. 11, pp. 3243–3257, 2022.

[67] H. Wang, W. Shi, and C.-S. Choy, "Hardware design of real time epileptic seizure detection based on STFT and SVM," *IEEE Access*, 2018.

[68] H. Elhosary, M. H. Zakhari, M. A. Elgammal, M. A. Abd El Ghany, K. N. Salama, and H. Mostafa, "Low-power hardware implementation of a support vector machine training and classification for neural seizure detection," *IEEE Transactions on Biomedical Circuits and Systems*, vol. 13, no. 6, pp. 1324–1337, 2019.

[69] K. Farhang Razi and A. Schmid, "Epileptic seizure detection with patient-specific feature and channel selection for low-power applications," *IEEE Transactions on Biomedical Circuits and Systems*, vol. 16, no. 4, pp. 626–635, 2022.

[70] A. Vittimberga, G. Nicolini, and G. Scotti, "Real-time multi-channel epileptic seizure detection exploiting an ultra-low-complexity algorithm–hardware co-design approach," *Sensors*, vol. 25, no. 22, p. 6889, 2025.

[71] P. Busia, G. Leone, A. Matticola, L. Raffo, and P. Meloni, "Wearable epilepsy seizure detection on FPGA with spiking neural networks," *IEEE Transactions on Biomedical Circuits and Systems*, vol. 19, no. 6, 2025.

[72] D. Sopic, A. Aminifar, and D. Atienza, "e-Glass: A wearable system for real-time detection of epileptic seizures," in *2018 IEEE International Symposium on Circuits and Systems (ISCAS)*, 2018.

[73] P. Busia, A. Cossettini, T. M. Ingolfsson, S. Benatti, A. Burrello, V. J. B. Jung, M. Scherer, M. A. Scrugli, A. Bernini, P. Ducouret, P. Ryvlin, P. Meloni, and L. Benini, "Reducing false alarms in wearable seizure detection with EEGformer: A compact transformer model for MCUs," *IEEE Transactions on Biomedical Circuits and Systems*, vol. 18, no. 3, pp. 608–621, 2024.

[74] V. Kartsch, G. Tagliavini, M. Guermandi, S. Benatti, D. Rossi, and L. Benini, "BioWolf: A sub-10-mW 8-channel advanced brain–computer interface platform with a nine-core processor and BLE connectivity," *IEEE Transactions on Biomedical Circuits and Systems*, vol. 13, no. 5, pp. 893–906, 2019.

[75] T. M. Ingolfsson, A. Cossettini, S. Benatti, and L. Benini, "Energy-efficient tree-based EEG artifact detection," in *44th Annual International Conference of the IEEE Engineering in Medicine & Biology Society (EMBC)*, 2022, pp. 3723–3728.

[76] I. Hubara, M. Courbariaux, D. Soudry, R. El-Yaniv, and Y. Bengio, "Binarized neural networks," in *Advances in Neural Information Processing Systems 29 (NIPS 2016)*, 2016, pp. 4107–4115.

[77] Y. Umuroglu, N. J. Fraser, G. Gambardella, M. Blott, P. Leong, M. Jahre, and K. Vissers, "FINN: A framework for fast, scalable binarized neural network inference," in *Proceedings of the 2017 ACM/SIGDA International Symposium on Field-Programmable Gate Arrays (FPGA '17)*. ACM, 2017, pp. 65–74.

[78] B. Jacob, S. Kligys, B. Chen, M. Zhu, M. Tang, A. Howard, H. Adam, and D. Kalenichenko, "Quantization and training of neural networks for efficient integer-arithmetic-only inference," in *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, 2018, pp. 2704–2713.

[79] J. Duarte, S. Han, P. Harris, S. Jindariani, E. Kreinar, B. Kreis, J. Ngadiuba, M. Pierini, R. Rivera, N. Tran, and Z. Wu, "Fast inference of deep neural networks in FPGAs for particle physics," *Journal of Instrumentation*, vol. 13, no. 07, p. P07027, 2018.

Appendix A

Full Results and Sources

A.1 All Models on the 29 Test Patients

Table A.1 Final evaluation of every model on the 29 test patients (5 seeds; SzCORE event metrics at the best-F1 operating point chosen on the training folds; Paper: the base paper's sensitivity and false alarms per hour; p: paired Wilcoxon on per-patient event-F1).

Model	AUROC	Event-F1	Sens	Prec	FA/24 h	Paper	Cost	p vs LGBM
DWN + distillation (final)	0.883	**0.248**	0.303	0.291	0.8	25.5 % / 0.02	895 LUTs	0.039
DWN + modality dropout	0.866	0.242	0.272	0.282	0.4	21.7 % / 0.01	951 LUTs	0.035
DWN base	0.875	0.172	0.340	0.156	2.7	29.2 % / 0.12	980 LUTs	0.004
LightGBM	0.877	0.308	0.331	0.396	0.6	26.8 % / 0.01	30,750 LUTs	–
XGBoost	0.872	0.294	0.301	0.387	0.4	25.1 % / 0.01	48,295 LUTs	0.29
CatBoost	0.874	0.254	0.330	0.281	1.2	25.8 % / 0.04	27,720 LUTs	0.026
Random forest	0.858	0.246	0.245	0.305	0.2	22.4 % / 0.00	346,576 LUTs	0.35
TreeLUT (5-bit leaves)	0.858	0.233	0.292	0.296	1.0	25.1 % / 0.04	6,330 LUTs	0.005
Logic-gate network	0.865	0.191	0.291	0.186	1.4	23.1 % / 0.06	7,452 LUTs	0.008
INT8 MLP	0.788	0.155	0.249	0.163	1.7	21.8 % / 0.10	9,600 mult.	0.005
Logistic regression	0.814	0.123	0.255	0.096	5.1	23.0 % / 0.27	117 mult.	0.0008
ChronoNet (clean, 1 seed)	–	0.027	0.790	0.015	106.3	67.1 % / 11.9	CNN + GRU	–

A.2 Sources of the Reported Numbers

Every number in this report comes from a saved result file; each run folder holds the configuration, code version, log and metrics.

Table A.2 Sources of the reported numbers.

Numbers	Source
Dataset inventory and audit (Chapter 3)	`reports/data_audit.md`, `results/audit/`
Fixed-point specification	`spec/fixed_point.md`
Integer vs floating point (Table 4.1)	`results/m3/int_float_gap.md`
ChronoNet reproduction (Figure 4.1)	runs `20261107-125159_chrononet_official_s1`, `20261107-204035_chrononet_clean_s1`
Model families (Table 4.2)	`results/tables/inner_cv.md`
Extensions, missing sensors (Figures 4.3, 4.5)	`results/tables/m5_ablation.md`; `breakdown.json` in each run folder
Final test (Tables 4.4, A.1)	`results/tables/final_test.md`
Base-paper comparison (Figure 4.9)	`results/tables/sens_fa_test.json`
Verilog check (Table 4.7)	`export/n4_distill_s0/verilog_check.json`
Decisions and their evidence	`results/decisions.md`